

TYUKALINSK, Russian Federation

WMO: 285860

Lat: 55.867N

Lon: 72.217E

Elev: 386

StdP: 14.49

Time Zone: 6.00 (E06)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.5	-24.9	-35.6	0.7	-30.1	-30.1	1.0	-24.7	18.5	12.8	17.1	13.1	2.6	100	0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	84.8	66.6	81.6	65.3	78.6	64.3	69.2	80.0	67.5	77.5	66.0	75.6	7.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	95.8	73.5	63.8	89.8	72.6	62.1	84.7	71.2	33.6	80.2	32.2	77.7	31.0	75.7	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	16.9	14.8	DB	-34.3	90.1	6.5	3.9	-39.0	92.9	-42.9	95.2	-46.5	97.4	-51.3	100.2
			WB	-34.2	71.7	6.9	1.9	-39.2	73.1	-43.2	74.2	-47.1	75.2	-52.1	76.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.7	-0.8	3.9	19.9	37.7	52.8	62.4	64.9	61.1	49.8	37.4	20.1	5.1
	DBStd	25.47	13.91	12.95	10.32	10.10	9.42	7.98	6.67	7.06	7.67	8.65	12.94	14.41
	HDD50	7057	1575	1292	933	384	81	8	0	3	94	397	898	1392
	HDD65	11269	2040	1712	1398	819	389	139	87	165	459	856	1348	1857
	CDD50	1469	0	0	0	15	169	381	463	346	89	6	0	0
	CDD65	206	0	0	0	0	12	62	85	43	4	0	0	0
	CDH74	1973	0	0	0	6	201	619	729	376	42	0	0	0
	CDH80	468	0	0	0	0	46	165	170	81	6	0	0	0
Wind	WSAvg	6.8	6.2	6.3	7.5	7.8	7.8	6.5	6.0	6.0	6.4	6.9	7.3	6.9
Precipitation	PrecAvg	16.4	0.6	0.5	0.6	1.0	1.5	2.3	2.8	2.4	1.5	1.2	1.1	0.9
	PrecMax	24.1	1.1	0.9	1.9	1.9	3.2	4.1	8.4	5.0	3.0	2.4	2.2	1.5
	PrecMin	11.9	0.0	0.0	0.0	0.1	0.4	1.0	0.8	0.9	0.4	0.3	0.3	0.3
	PrecStd	3.1	0.3	0.2	0.4	0.5	0.8	0.8	1.7	1.0	0.7	0.6	0.5	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.3	34.1	43.5	72.7	85.1	88.4	88.8	86.1	79.8	63.1	43.6	34.9
		MCWB	29.7	31.2	38.0	54.2	60.6	65.6	69.3	67.4	60.9	50.7	40.9	33.4
	2%	DB	27.9	30.9	39.1	64.0	79.1	84.5	84.4	81.9	72.2	58.8	40.4	31.8
		MCWB	26.8	29.0	35.0	50.4	59.3	66.4	67.7	66.2	57.0	48.9	38.1	30.4
	5%	DB	24.6	27.6	36.2	58.8	74.8	80.8	81.0	78.2	67.1	54.2	37.6	28.4
		MCWB	23.7	26.2	33.2	48.0	57.7	65.0	66.4	64.6	56.1	46.6	35.4	26.9
	10%	DB	20.2	23.3	33.8	53.6	70.0	77.1	78.0	74.3	63.0	50.2	34.7	24.8
		MCWB	19.2	22.0	31.4	44.9	55.3	63.7	65.5	63.2	53.9	44.8	33.0	23.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.2	31.6	39.2	55.8	63.9	71.4	71.7	70.2	62.9	53.4	41.1	33.7
		MCDB	31.2	33.3	42.3	69.3	77.4	82.0	82.7	81.3	74.3	60.1	42.8	34.6
	2%	WB	26.8	29.1	35.6	51.4	61.0	68.3	69.6	67.9	59.5	50.3	38.6	30.7
		MCDB	27.8	30.6	38.2	62.1	75.5	79.0	80.6	78.3	68.4	56.5	40.2	31.7
	5%	WB	23.7	26.2	33.5	48.5	58.6	66.6	68.1	66.1	57.1	47.6	35.5	27.0
		MCDB	24.6	27.6	35.8	57.7	72.1	77.3	77.8	75.1	65.5	53.0	37.2	28.4
	10%	WB	19.3	22.0	31.5	45.6	56.5	64.7	66.6	64.2	54.5	45.0	33.2	23.6
		MCDB	20.2	23.3	33.8	53.2	68.3	75.0	75.8	72.3	62.1	50.0	34.5	24.8
Mean Daily Temperature Range	5% DB	MDBR	13.8	16.0	16.1	16.3	20.0	18.4	17.4	17.4	15.9	12.4	10.5	13.4
		MCDBR	13.7	12.5	14.3	25.2	27.5	24.7	22.4	24.2	24.0	20.1	10.4	11.7
		MCWBR	13.5	11.8	11.6	14.6	12.2	10.6	9.1	10.9	12.6	12.2	9.2	11.5
	5% WB	MCDBR	13.4	12.4	13.2	23.6	25.2	21.6	19.7	20.5	21.3	17.3	9.9	11.7
		MCWBR	13.3	11.8	11.2	14.3	12.5	10.3	9.3	10.3	12.4	11.9	9.3	11.5
Clear-Sky Solar Irradiance	taub	0.278	0.312	0.368	0.412	0.404	0.407	0.429	0.430	0.371	0.354	0.315	0.283	
	taud	2.043	2.021	1.939	2.085	2.233	2.295	2.229	2.193	2.323	2.222	2.090	1.990	
	Ebn at Noon	189	229	243	253	265	266	256	246	246	214	170	149	
	Edh at Noon	21	31	44	44	41	39	41	39	30	26	20	17	
All-Sky Solar Radiation	RadAvg	276	612	1105	1460	1728	1857	1728	1365	930	476	231	175	
	RadStd	21	32	77	112	141	134	127	120	90	56	37	21	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page